

plot_trait_group_distribution.R

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plot_trait_group_distribution
<i>Plot Raw Trait Value Distribution for One Taxon-Domain Group</i>

Description

Builds a ‘ggplot2’ strip + boxplot showing every raw trait value for a single ‘taxon_name x trait_domain_name’ group, with colour-coded Tukey fence flags and horizontal fence lines for both the standard (1.5x IQR) and extreme (3x IQR) thresholds.

Usage

```
plot_trait_group_distribution(  
  data_group_raw,  
  data_group_summary,  
  sel_taxon,  
  sel_domain,  
  graphical_options,  
  verbose = TRUE  
)
```

Arguments

- | | |
|--------------------|---|
| data_group_raw | A tibble of raw trait observations for one taxon x domain group. Must contain columns ‘trait_value’ (numeric), ‘trait_name’ (character), and ‘trait_domain_name’ (character). |
| data_group_summary | A single-row tibble of per-group QC statistics, as produced by ‘generate_trait_qc_report()’. Must contain numeric columns ‘mean’, ‘median’, ‘IQR’, and integer columns ‘n_suspected_outliers_taxon’ (integer) and ‘outlier_fraction’ (numeric). |
| sel_taxon | Character scalar. Name of the taxon being inspected. Used in the plot title. |

<code>sel_domain</code>	Character scalar. Name of the trait domain being inspected. Used in the plot title.
<code>graphical_options</code>	Named list with elements <code>'width'</code> , <code>'height'</code> , <code>'units'</code> , <code>'dpi'</code> , and <code>'bg'</code> , as returned by <code>'get_active_config("graphical")'</code> . Passed to <code>'ggview::canvas()'</code> .
<code>verbose</code>	Logical. If <code>'TRUE'</code> (default), the computed Tukey fence boundaries are printed to the console via <code>'cli::cli_inform()'</code> .

Details

The function computes Q1, Q3, and IQR from `'data_group_raw'` and derives four fence values: - inner lower / upper: $Q1 - 1.5 * IQR$ and $Q3 + 1.5 * IQR$ - outer lower / upper: $Q1 - 3 * IQR$ and $Q3 + 3 * IQR$

Each observation is classified as `"within fence"`, `"mild outlier (1.5x IQR)"`, or `"extreme outlier (3x IQR)"`. When `'data_group_raw'` contains more than one distinct `'trait_name'`, observations are faceted by `'trait_name'`; otherwise the x-axis shows `'trait_domain_name'`.

Plot layer order follows project conventions: `'ggplot()'` -> `facets` -> `scales` -> `labels` -> `theme` -> `'ggview::canvas()'` -> `geoms`.

Value

A `'ggplot2'` object. The plot is not printed; call `'print()'` on the return value to display it.

See Also

`[generate_trait_qc_report()]`, `[apply_trait_corrections()]`, `[validate_trait_corrections()]`

Examples

```
## Not run:
data_raw <-
  data_traits_raw |>
  dplyr::filter(
    taxon_name == "Anacyclus clavatus",
    trait_domain_name == "Leaf Area"
  ) |>
  dplyr::arrange(trait_value)

data_summary <-
  data_qc_report |>
  dplyr::filter(
    taxon_name == "Anacyclus clavatus",
    trait_domain_name == "Leaf Area"
  )

graphical_options <-
  get_active_config("graphical")

p <-
  plot_trait_group_distribution(
    data_group_raw = data_raw,
    data_group_summary = data_summary,
    sel_taxon = "Anacyclus clavatus",
    sel_domain = "Leaf Area",
    graphical_options = graphical_options
```

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)

base::print(p)

End(Not run)

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